

DECODELABS

DATA ANALYTICS INTERNSHIP

Project 1: Data Cleaning & Preparation

Official Data Integrity Artifact & Audit Log

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Domain:	Data Analytics
Submission Date:	June 2026
Dataset:	Dataset for Data Analytics.xlsx

1. Executive Summary

This document serves as the official data integrity artifact for **Project 1: Data Cleaning & Preparation**. A comprehensive audit was conducted on a transactional dataset containing **1,200 records** and **14 columns**. The objective was to identify missing values, verify record uniqueness, validate data formats, and assess business logic consistency.

The audit identified **309 missing values** in the **CouponCode** column. These values were determined to represent transactions where no coupon was applied and were replaced with the standard value **"NO_COUPON"**. Additional validation checks confirmed that the dataset contained no duplicate records, no duplicate Order IDs, no date inconsistencies, and no business logic violations.

The dataset successfully passed all validation checks and is **ready for downstream analysis and reporting**.

2. Dataset Overview

Total Records: 1,200

Total Columns: 14

Columns Audited:

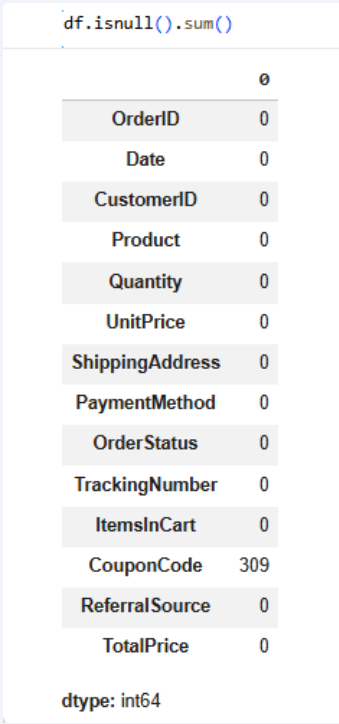
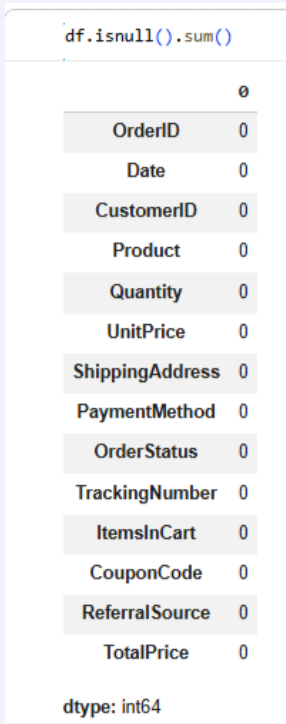
- OrderID
- CustomerID
- Quantity
- ShippingAddress
- OrderStatus
- ItemsInCart
- ReferralSource
- Date
- Product
- UnitPrice
- PaymentMethod
- TrackingNumber
- CouponCode
- TotalPrice

3. Data Transformation Audit Log

Change ID	Column Name	Description of Transformation	Business Impact & Justification	Status
CR001	CouponCode	Identified 309 missing values and replaced them with "NO_COUPON".	Eliminated missing values while preserving all transaction records. Improved data completeness for analysis and reporting.	Resolved
CR002	OrderID	Performed duplicate record validation.	Confirmed all Order IDs are unique, ensuring transaction integrity and preventing duplicate reporting.	Verified
CR003	Date	Validated date formatting and consistency.	Confirmed all records contain valid and standardised date values for time-based analysis.	Verified
CR004	TotalPrice	Verified TotalPrice against Quantity × UnitPrice.	Confirmed transaction calculations are accurate and reliable.	Verified
CR005	Entire Dataset	Performed data type, null value, categorical consistency, and data quality checks.	Confirmed dataset readiness for downstream analytics and reporting.	Verified

3a. Verification Screenshots — Null Value Check

The screenshots below show the output of `df.isnull().sum()` before and after the CouponCode cleaning step, confirming that 309 missing values were successfully identified and resolved.

<pre>df.isnull().sum()</pre>  <pre> 0 OrderID 0 Date 0 CustomerID 0 Product 0 Quantity 0 UnitPrice 0 ShippingAddress 0 PaymentMethod 0 OrderStatus 0 TrackingNumber 0 ItemsInCart 0 CouponCode 309 ReferralSource 0 TotalPrice 0 dtype: int64 </pre> BEFORE Cleaning CouponCode = 309 missing values	<pre>df.isnull().sum()</pre>  <pre> 0 OrderID 0 Date 0 CustomerID 0 Product 0 Quantity 0 UnitPrice 0 ShippingAddress 0 PaymentMethod 0 OrderStatus 0 TrackingNumber 0 ItemsInCart 0 CouponCode 0 ReferralSource 0 TotalPrice 0 dtype: int64 </pre> AFTER Cleaning CouponCode = 0 missing values
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4. Audit Summary

	Validation Check	Result
✓	Missing Values	309 identified and resolved in CouponCode
✓	Duplicate Rows	0 Found
✓	Duplicate Order IDs	0 Found
✓	Invalid Date Formats	0 Found
✓	Data Type Issues	0 Found
✓	Business Logic Violations	0 Found
✓	Dataset Readiness	Passed

5. Verification Gate Conclusion

The dataset has successfully passed all data quality validation checks. Missing values were identified, assessed, and appropriately handled. Duplicate records were not present, date fields were consistent, and transaction calculations were verified as accurate.

✓ DATASET STATUS: ANALYSIS READY

The dataset is approved for subsequent reporting, visualisation, and business intelligence activities.